

Education

The Pennsylvania State University
Bachelor of Science: Computer Science Major

Expected Graduation: Spring 2025
Harrisburg, PA

- Honors: **Dean's List, Bunton-Waller Scholarship recipient**

GPA 3.3

Relevant Experience

Phoenix Contact

Technician

May 2021 – Present

Middletown, PA

- Created interactive SOP references by training AI models on internal documentation, improving on-floor accessibility and reducing human error.
- Led informal training for coworkers on using AI tools to enhance day-to-day technical operations.
- Interpreted blueprints and configured machine setups, optimizing production efficiency and reducing setup time.
- Utilized AI tools like ChatGPT and Python scripts to assist in interpreting technical documentation and generating automated troubleshooting workflows.
- Performed high-precision PCB soldering to ensure product reliability and compliance with quality standards.

Projects

Clarify Task Management App

React Native | Firebase | Firestore

- Developed a productivity-focused task management app, Clarify, using React Native for mobile platforms to help users organize tasks effectively.
- Integrated Firebase and Firestore for real-time data storage and synchronization, ensuring seamless user experience across devices.
- Implemented custom themes, notifications, and recurring tasks for enhanced usability and personalization.

Machine Learning-Assisted Rodent Video Analysis Tool [Capstone Final Project]

Python | OpenCV | SQL | Django

- Combined OpenCV & DeepLabCut for real-time pose tracking in behavioral studies
- Built a Django+SQL backend for high-throughput data analysis
- Engineered data preprocessing pipelines and integrated statistical analysis to extract behavioral metrics, enabling more accurate and automated insight generation for neuroscience research

Compiler Construction Toolkit: Syntax Sorcerer

Eclipse | IntelliJ | Jflex | Java

- Developed a lexical analyzer using JFlex that tokenizes Java source code based on custom regular expression rules.
- Built a Java-based recursive descent parser to construct structured syntax trees, enabling syntactic analysis of programming constructs
- Implemented error handling mechanisms for invalid tokens and syntax errors to provide real-time feedback during compilation.

Transaction Data Management System (TDMS)

Python | MySQL | SQL

- Designed and implemented a star-schema database architecture using partitioned fact tables and dimension tables for optimized OLAP performance.
- Integrated real-time transaction data ingestion pipelines to support timely analysis and anomaly tracking.
- demonstrated proficiency in SQL performance tuning, schema design, and scalable data architecture

Car Service Management Application

Java | MySQL | SQL | JDBC | IntelliJ

- Developed a Java-based desktop application simulating a car rental service with dual platforms: user/renter and admin.
- Designed and implemented the MySQL database schema to manage vehicles, bookings, users, and admin control functions.
- Integrated features such as booking management, vehicle availability, and administrative approval workflows.

Honors & Awards

Third Place in Programming Competition (April 2023): Distinguished myself by securing third place in a pioneering AI-driven programming competition, which underscored my adept utilization of advanced AI tools such as ChatGPT in the resolution of intricate algorithmic challenges.

Technical Skills

Languages: C++, Java, Python, JavaScript, SQL, HTML, CSS

Frameworks & Tools: Flask, Firebase, React, Vercel, GitHub, Azure, Jflex

Software & IDEs: Microsoft Office, G Suite, Git, VS Code, IntelliJ, PyCharm, CLion